

Simulation of a smart warehouse monitoring system

1.Simulating sensors using in Python:

The 5 things that should be monitored in a warehouse in order to ensure the safety of the objects are:

1. Temperature: Goods such as medicine are affected by rising temperatures.
2. Humidity: Enhances mold growth which can cause material corrosion.
3. Vibration: used in predictive maintenance by indicating abnormal material behaviour.
4. Pallet presence: Used for storage visibility and inventory tracking.
5. Battery level: batteries allow predictive maintenance.

Things to consider when simulating value from a sensor:

1. Initial values
2. Sensor data fluctuations
3. Range
4. Effect of low battery
5. Rounding of values for better readings
6. Random sensor failures
7. Sampling rate
8. Converting data lastly to Json for being lightweight and a better structure

Code: https://github.com/SalBarakat/smart-monitoring-warehouse-system/blob/832f0a1950dd33ee66faff830d269ba3577b4e26/code_smart_warehouse.py

2.Add MQTT Layer:

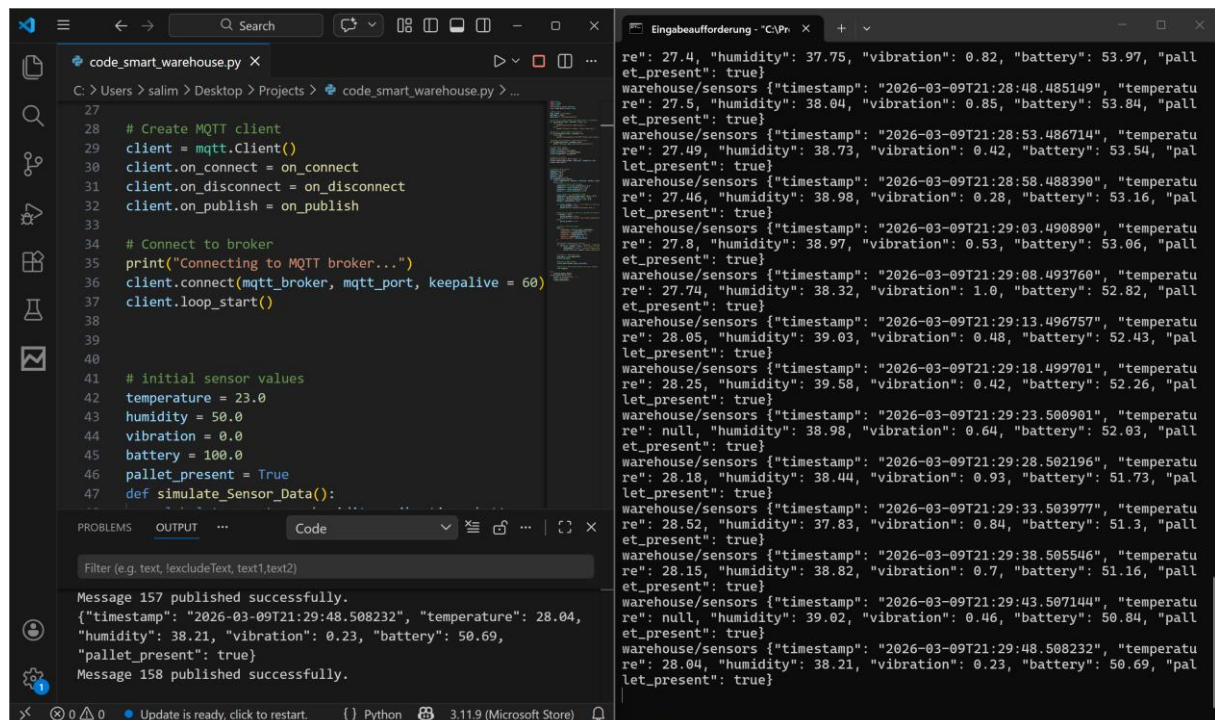
MQTT is a lightweight protocol which allows communication between IoT devices and clients. This makes it possible to monitor all the readings of different sensors from a central system in the warehouse office.

Here I used MQTT as a local host and this video explains how to set it up:

<https://youtu.be/2zzUIPHKtMk?si=LQYrXCjEtHrJxArP>

- You have to install Mosquitto and add “pip install paho-mqtt” in your terminal.
- Next run the code and you should get the message: “MQTT is connected”
- Now paste this code in your command prompt: "C:\Program Files\mosquitto\mosquitto_sub" -h localhost -t "**your_topic_name**" -v

Doing those steps should show your console's output in the command prompt



The screenshot shows a VS Code editor with a Python file named `code_smart_warehouse.py`. The script creates an MQTT client, connects to a local broker, and publishes sensor data. The console output shows the connection status and the published messages.

```
27 # Create MQTT client
28 client = mqtt.Client()
29 client.on_connect = on_connect
30 client.on_disconnect = on_disconnect
31 client.on_publish = on_publish
32
33 # Connect to broker
34 print("Connecting to MQTT broker...")
35 client.connect(mqtt_broker, mqtt_port, keepalive = 60)
36 client.loop_start()
37
38 # initial sensor values
39 temperature = 23.0
40 humidity = 50.0
41 vibration = 0.0
42 battery = 100.0
43 pallet_present = True
44 def simulate_Sensor_Data():
45     ...
```

Console Output:

```
Message 157 published successfully.
{"timestamp": "2026-03-09T21:29:48.508232", "temperature": 28.04,
"humidity": 38.21, "vibration": 0.23, "battery": 50.69,
"pallet_present": true}
Message 158 published successfully.
```

3.Adding backend and database:

For the database I chose to use SQLite since it is easy and good for prototyping, but for a professional system it is better to use another tool.

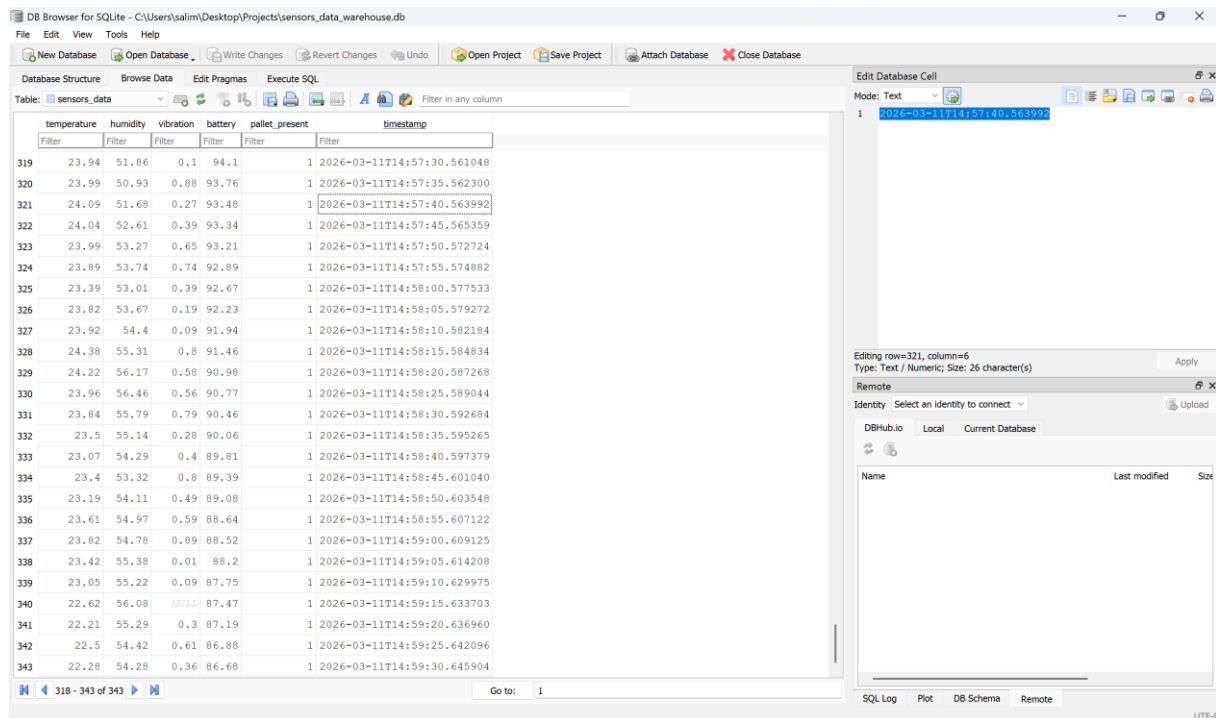
Here is a YouTube playlist that helped me understand the fundamentals of SQLite:

https://youtube.com/playlist?list=PLJJcOjd3n1ZfwZKYM_uJSs0xSejPsQlav&si=TLRZfFrGKizUtO76

write in another file a new python script for saving the data in an SQL database.

Code: <https://github.com/SalBarakat/smart-monitoring-warehouse-system/blob/832f0a1950dd33ee66faff830d269ba3577b4e26/database.py>

Then run the new file and the other file in two different terminals.



A mistake that I got: I opened another SQL file with the same name but couldn't find any data even after refreshing the file.

Solution: write this code below in your script and then copy the path in your file explorer to find the right SQL file.

```
import os
```

```
print(f"REAL DATABASE LOCATION: {os.path.abspath('warehouse.db')}")
```

replace warehouse.db with the name of your file that you are looking for.

4. Dashboard (Flask):

Flask is dynamic, unlike HTML. So, Flask can change whenever you refresh the page and show the newest sensor data.

An introduction video to Flask that have helped me: (you don't have to watch the whole video, since it isn't really relevant to this particular project)

https://youtu.be/Z1RJmh_OqeA?si=QboqooNSZl9JmZnW

Write a third script for showing the data in a dashboard.

First, we have to download the Flask libraries using “pip install flask”

Also “pip install paho-mqtt flask flask-sqlalchemy”

As well as another fourth script of HTML

However, you have to create a folder from the start in which you’ll add all of your relevant codes for your project and inside it create a new folder and add your HTML code in that folder.

Flask Code: <https://github.com/SalBarakat/smart-monitoring-warehouse-system/blob/832f0a1950dd33ee66faff830d269ba3577b4e26/app.py>

HTML code: <https://github.com/SalBarakat/smart-monitoring-warehouse-system/blob/832f0a1950dd33ee66faff830d269ba3577b4e26/index.html>

Possible problem: HTML code in VS code isn’t working, because code language isn’t supported or defined.

Solution: run your app code, the file with the flask code and then the HTML code will automatically run.

Run the sensor simulating file, the database file and the flask code each in a different terminal and open the localhost in which a website link will be given to you in the output after running the flask file.

